THE CONNECTED COMPONENT ALONG A SECTION IS LOCALLY CONSTRUCTIBLE FOR A FINITELY PRESENTED MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8267-8273 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 8275-8307, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Connected components along a section, constructibility of finite-type loci, and finite presentation of the morphism; the analysis is retained.

Source reference. Original lemma lemma-base-change-connected-along-section: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8187–8199. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-connected-along-section-good: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8218–8225. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8307–8307. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8307–8307. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma situation-connected-along-section: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8179–8184. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 053Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 8307–8307. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$, $s : Y \to X$ be as in Situation situation-connected-along-section. If f is of finite presentation then X^0 is locally constructible in X.*

Proof. Let $x \in X$. We have to show that there exists an open neighbourhood U of x such that $X^0 \cap U$ is constructible in U. This reduces us to the case where Y is affine. Write $Y = \mathrm{Spec}(A)$ and $A = \mathrm{colim}\, A_i$ as a directed limit of finite type $\mathbf{Z}$-algebras. By Limits, Lemma 01ZM we can find an i and a morphism $f_i : X_i \to \mathrm{Spec}(A_i)$ of finite presentation, endowed with a section $s_i : \mathrm{Spec}(A_i) \to X_i$ whose base change to Y recovers f and the section s. By Lemma lemma-base-change-connected-along-section it suffices to prove the lemma for f_i, s_i. Thus we reduce to the case where Y is the spectrum of a Noetherian ring.

Assume Y is a Noetherian affine scheme. Since f is of finite presentation, i.e., of finite type, we see that X is a Noetherian scheme too, see Morphisms, Lemma 01T6. In order to prove the lemma in this case it suffices to show that for every irreducible closed subset $Z \subset X$ the intersection $Z \cap X^0$ either contains a nonempty open of Z or is not dense in Z, see Topology, Lemma 053Z. Let $x \in Z$ be the generic point, and let $y = f(x)$. By Lemma lemma-connected-along-section-good there exists a nonempty open subset $V \subset \overline{\{y\}}$ such that $X^0 \cap X_V$ is open and closed in X_V. Since $f(Z) \subset \overline{\{y\}}$ and $f(x) = y \in V$ we see that $W = f^{-1}(V) \cap Z$ is a nonempty open subset of Z. It follows that $X^0 \cap W$ is open and closed in W. Since W is irreducible we see that $X^0 \cap W$ is either empty or equal to W. This proves the lemma. $\square$